

Dhairya Joshi

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SUMMARY

B.Tech student at IIT Jodhpur (Minor in AI) specializing in **machine learning, deep learning, LLMs, and AI systems**. Experienced in **LLM fine-tuning, multi-GPU inference, AI guardrails, ONNX deployment, async systems, and quantitative ML**, with hands-on experience building end-to-end production-oriented systems.

EDUCATION

Indian Institute of Technology Jodhpur

Jodhpur, India

B.Tech in Mechanical Engineering (Minor in AI), CGPA: 8.43

2024 – 2028

- Coursework: Data Structures & Algorithms (A) | Machine Learning (A-) | Scientific Computations (A-) | Deep Learning | Probability & Statistics | Natural Language Understanding (NLU) | Mathematics

TECHNICAL SKILLS

Languages: Python, C++, SQL, Rust

Machine Learning & Deep Learning: Scikit-Learn, Random Forest, XGBoost, Gradient Boosting, Feature Engineering, PyTorch, TensorFlow, Keras, CNN, RNN, LSTM, GRU, GANs, Transformer Architectures (Llama, Gemma, BERT, DeBERTa)

Generative AI / LLMs & Systems: LLM Fine-Tuning, RAG, LangChain, Vector Databases, Prompt Engineering, LLM Guardrails & Safety Pipelines, FastAPI, REST APIs, ONNX Runtime, Async Concurrency, Multi-GPU Pipelined Inference

Data Science: NumPy, Pandas, Matplotlib

Tools: Git, GitHub

EXPERIENCE

Undergraduate Research Assistant

Dec. 2025 – Jan. 2026

Supervisor: Prof. G. Venkat Ram Reddy, SME, IIT Jodhpur

- Automated finance payment processing with **Playwright and n8n**, enforcing human-in-the-loop approval gates for zero unauthorized executions across 10 autonomous payment runs.
- Integrated **event-driven webhooks** with fallback sweeps and a JSONL audit trail, enabling autonomous recovery of missed events and sub-15s approval resolution with duplicate-free records.
- Optimized a **log-segmentation model on 162K events** to partition operation logs into business-process executions, validated on 20K unseen events, cutting per-operation processing time **10x (20s → 2s)**.

PROJECTS

CheckerPlane AI – Low-Latency LLM Guardrail Proxy

GitHub

July 2026 – Aug. 2026

Rust, ONNX Runtime, C++, DeBERTa, BERT, Tokio, Async Concurrency

- Architected a **multi-tier AI safety proxy in Rust** with ONNX Runtime, replacing sequential Python guardrail chains with parallel async execution – cutting median latency **~8x (550ms → 67ms)**.
- Implemented a 4-gate parallel input rail (toxicity, PII/Aadhaar NER, coherence, prompt-injection) with early-exit cancellation and a **<0.1ms Tier-0 heuristic screen**.
- Trained a **DeBERTa NLI hallucination gate** gate scoring output grounding, reaching ~88% gate accuracy with zero GPU dependency via quantized ONNX models.

Delta Hedging & Adversarial Stock Path Generation Simulator

GitHub

May 2026 – June 2026

Python, TensorFlow, NumPy, Pandas, Monte Carlo Simulation, Black-Scholes Model

- Built a quant finance simulator benchmarking classical delta hedging against deep learning-based strategies across 10,000+ Monte Carlo-simulated market scenarios.
- Constructed a **GRU-based hedging model** with a custom CVaR loss function, cutting tail risk (CVaR_{5%}) 42% versus the classical baseline (Rs. 500 → Rs. 290).
- Designed an **adversarial framework** where a recurrent generator learns worst-case market conditions, pushing the tail-risk reduction to 80% overall (Rs. 100).

LLM Response Preference Classification – Kaggle Competition

GitHub

Aug. 2026 – Sept. 2026

Python, PyTorch, HuggingFace Transformers, Llama 3, Gemma 2, xFormers, Multi-GPU Inference

- Fine-tuned Llama-3 and Gemma-2 classifiers on ~57,400 datapoints to predict human preference between LLM response pairs (win A / win B / tie).
- Engineered a **pipelined multi-GPU inference system** splitting Llama-3 across two GPUs, using xFormers block-diagonal attention and sharded batching to eliminate padding waste.
- Benchmarked against an LSTM baseline, cutting log-loss from 1.13 to 0.83 (~27%), motivating the switch to transformer-based classification.
- Ensembled predictions with test-time augmentation for a 0.83 log-loss – a top-1% finish among 224 teams.

ACHIEVEMENTS

- **Ranked 2nd of 224 teams (top 1%)** in the Kaggle ‘LLM Classification Finetuning’ competition, competing solo.
- **Secured 1st Rank** in the Prometeo ML CodeFest by developing a Delta Hedging & Adversarial Stock Path Generation Simulator for quantitative finance.